

PEER: Unified Process–Outcome Reinforcement Learning for Structured Empathetic Reasoning

Yunxiao Wang, Meng Liu, Sicheng Zhao, Lizi Liao, and Liqiang Nie

Abstract—Emotional support conversations require more than fluent responses. Supporters need to understand the seeker’s situation and emotions, adopt an appropriate strategy, and respond in a natural, human-like manner. Despite advances in large language models, current systems often lack structured, psychology-informed reasoning. Additionally, it is challenging to enhance these systems through reinforcement learning because of unreliable reward signals. Moreover, reinforcement fine-tuning can amplify repetitive response patterns. We propose *structured empathetic reasoning*, which breaks support into three steps: conversation history analysis, multimodal emotional state inference, and strategy selection, prior to generating the final reply. To implement this, we introduce SER, a fine-grained dataset with step-level correctness labels and pairwise response preferences. We then present PEER, which uses GRPO with *UnifiReward*, a unified process–outcome reward model for evaluating both reasoning steps and final responses in multi-turn interactions. To reduce repetition, we enhance data with personality-based rewriting and down-weight redundant outputs. Comprehensive experiments show improved empathy, strategy alignment, and human-likeness without sacrificing diversity. Code and data are available at <https://github.com/Yunxiao-Wang/PEER>.

Index Terms—Emotional Support Conversations, Psychology-Informed Reasoning, Reinforcement Learning, Reward Model.

I. INTRODUCTION

Emotional Support Conversation (ESC) aims to comfort people experiencing emotional distress through natural dialogue [1]. With Large Language Models (LLMs), ESC systems have improved substantially in fluency and general helpfulness [2]. Recently, visual information, including facial expressions, has also been incorporated to achieve multimodal ESC [3], [4]. However, producing *effective* emotional support still requires more than writing a polite response. A supporter must infer the seeker’s situation and emotional state, choose an appropriate support strategy, and respond in a human-like manner. In practice, even strong LLMs frequently default to generic empathy, premature advice, or strategy-inconsistent responses, which limits their usefulness in real support settings.

A central reason is that ESC relies on higher-order social cognition rather than formal logical reasoning. Recent reasoning-oriented training has improved LLM performance on structured tasks [5], but these gains do not reliably transfer



Fig. 1. An illustration of structured empathetic reasoning. Given the seeker’s utterance together with the facial image, the supporter performs three reasoning steps before replying: (i) conversation history analysis to identify the seeker’s issue and the current support stage, (ii) multimodal emotional state inference from both facial expressions and text, and (iii) strategy selection. It then generates the final empathetic response conditioned on these steps.

to psychologically grounded dialogue behaviors [6]. Existing ESC methods [3], [7] partially address this gap by adding emotion annotations or prompting models to produce rationales. However, two limitations still persist. First, the supervision needed to learn such reasoning is missing from existing ESC corpora. They record only the final reply, without intermediate reasoning steps or labels on whether a step is correct. Trained on such data, standard supervised fine-tuning never observes *why* a response is appropriate. Second, optimizing the whole reasoning trajectory with Reinforcement Learning (RL) raises two concrete difficulties. The reward must judge the intermediate steps and the final response without the two judgments conflicting, and reward maximization itself erodes diversity (entropy collapse) [8], [9], pushing replies toward repetitive templates.

As shown in Fig. 1, we address these challenges with *structured empathetic reasoning*, a lightweight reasoning scaffold that decomposes emotional support into three steps before generating the final response: (i) *Conversation history analysis* to identify the seeker’s issue and the current support stage, (ii) *Multimodal emotional state inference* to assess affect based on both facial expressions and textual utterances, and (iii) *Strategy selection* to determine the appropriate response. This structure makes the model’s supportive behavior more interpretable and provides natural supervision targets beyond the final utterance.

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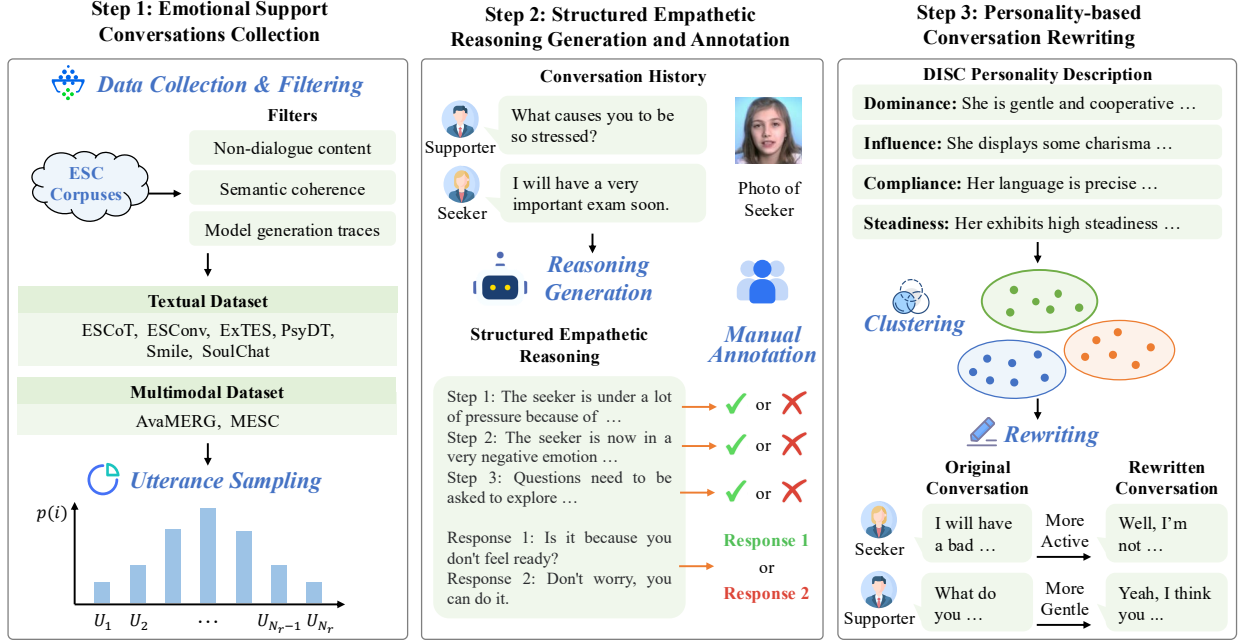


Fig. 2. Construction pipeline of the SER dataset. The pipeline comprises three stages: (i) emotional support conversation collection, where open-source corpora are gathered, filtered with an LLM, and sparsely sampled; (ii) structured empathetic reasoning generation and annotation, where candidate reasoning steps and responses are generated and then manually labeled with step-level correctness and pairwise response preferences; and (iii) personality-based conversation rewriting, where dialogues are rewritten with DISC-derived personas and filtered to enrich stylistic diversity.

To close this data gap, we first construct the **Structured Empathetic Reasoning (SER)** dataset, which is fine-grained annotated with (i) step-level correctness labels for the three reasoning steps and (ii) pairwise response preferences for the final reply. These annotations provide learning signals for both the reasoning process and response quality. Because diversity is crucial in ESC, we further augment SER with personality-based conversation rewriting, which increases stylistic coverage while preserving the original supportive intent.

To address the two difficulties of RL, we then propose the **emPathetic rEinforcEmEnt Reasoning (PEER)** framework that improves both reasoning fidelity and response quality. PEER uses GRPO [10] as the underlying optimizer and introduces *UnifiReward*, a **unified process–outcome reward model** that jointly evaluates intermediate reasoning steps (process) and the final response (outcome) in multi-turn settings. Unifying these signals within a single reward model diminishes the inconsistencies that occur when separate process and outcome evaluators reach different conclusions. Finally, to mitigate RL-induced repetition, we incorporate a **redundancy-aware reward reweighting** mechanism that down-weights overly similar candidate outputs relative to the conversation history and peer generations. Experiments on ESC benchmarks and human preference evaluation show that PEER improves empathy, strategy alignment, and human-likeness while maintaining response diversity.

Our contributions can be summarized as follows:

- We build SER, an ESC dataset annotated for structured empathetic reasoning, providing step-level correctness labels, pairwise response preferences, and personality-based rewriting for stylistic diversity.
- We propose PEER, which fine-tunes the policy on SER

and then optimizes it with GRPO under UnifiReward, a unified process–outcome reward model, together with redundancy-aware reward reweighting against entropy collapse.

- Experiments on ESC-Eval, SAGE, reference-based metrics, and human evaluation show that PEER outperforms general-purpose and emotion-specialized baselines while maintaining response diversity.

II. RELATED WORK

A. Emotional Support Conversation

ESC plays a crucial role in advancing the application of LLMs in fields such as psychological counseling and social forums. Liu et al. [1] introduced the first ESC dataset, ESCConv, by creating manually crafted conversations, which significantly promoted research on ESC systems. Meanwhile, Chu et al. [3] incorporated visual information, such as facial expressions, to support multimodal ESC. However, the manual construction of ESC datasets is costly and labor-intensive, resulting in limited dataset sizes. To address this limitation, recent studies have focused on leveraging LLMs to expand conversations based on real single-turn psychological counseling reports [7] or character-based prompts [11], [12], thereby generating larger datasets. Nevertheless, the dialogues generated by LLMs often lack stylistic flexibility and deep reasoning. ESCoT [13] incorporates a chain-of-thought approach to enhance the model's emotional support capabilities. However, it lacks annotations for reasoning correctness and outcome preference, which limits its applicability in reinforcement learning frameworks. We address this issue by refining the structure of the thought chain and providing manual annotations for both the reasoning

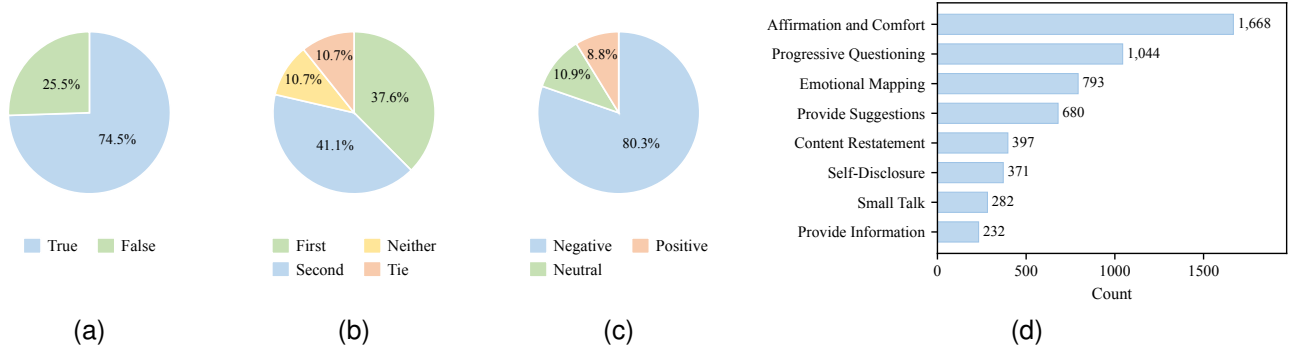


Fig. 3. Statistics of the SER dataset. (a) Distribution of the binary correctness labels of the reasoning steps; (b) distribution of the pairwise response-preference labels; (c) distribution of sentiment polarity; and (d) distribution over the support-strategy categories.

process and the final outcomes. Furthermore, the majority of existing works [3], [11], [14], [15] solely utilize supervised fine-tuning during model training. This training approach alone cannot fully exploit the potential of existing models and datasets. As a result, we incorporate reinforcement learning algorithms [16], [17] to train the model and optimize it for emotional support scenarios.

B. RL with Reward Models

In LLMs, RL was first widely used for human preference alignment, most notably via Reinforcement Learning from Human Feedback (RLHF) [18]. To simplify RLHF, DPO [19] bypasses reward modeling and directly optimizes the policy from preference pairs, improving efficiency and stability. More recently, RL has been applied to enhance LLM reasoning [5]. For example, GRPO [10] is designed to strengthen reasoning performance through group-based relative advantage estimation, which helps alleviate the effects of noisy or sparse reward signals and is particularly beneficial for challenging reasoning tasks such as mathematical problem solving. However, recent studies [8], [9], [20] show RL training can cause entropy collapse, pushing outputs toward overly rigid patterns.

Reward models play a central role in shifting LLM training from passive learning on static datasets to active optimization under dynamic feedback [21]. From the perspective of reward granularity, existing reward models can be broadly categorized into Outcome Reward Models (ORMs), which assess the quality of final answers, and Process Reward Models (PRMs), which evaluate intermediate reasoning steps in a fine-grained manner. These two paradigms provide complementary supervision signals: ORMs emphasize task-level correctness, whereas PRMs offer richer guidance over the reasoning trajectory. Their integration therefore has the potential to provide a more comprehensive reward signal and to further improve reasoning quality. Nevertheless, combining them also introduces new challenges, particularly when process-level assessments and final-outcome evaluations are not fully consistent [22].

III. OUR SER DATASET

A. Conversation Collection

As shown in Fig. 2, we first compile a comprehensive collection of open-source emotional support conversation

TABLE I
PROPERTY COMPARISON OF ESC DATASETS. MM: MULTIMODAL INPUT; RS: REASONING STEPS; SC: STEP-LEVEL CORRECTNESS LABELS; RP: PAIRWISE RESPONSE PREFERENCES.

Dataset	Lang.	MM	RS	SC	RP
ESConv [1]	En	×	×	×	×
MESC [3]	En	✓	×	×	×
AvaMERG [23]	En	✓	✓	×	×
ExTES [24]	En	×	×	×	×
ESCoT [13]	En	×	✓	×	×
SoulChat [15]	Zh	×	×	×	×
PsyDTCorpus [11]	Zh	×	×	×	×
CPsyCoun [14]	Zh	×	×	×	×
SmileChat [2]	Zh	×	×	×	×
SER (Ours)	En+Zh	✓	✓	✓	✓

datasets, listed in TABLE I. These datasets cover both Chinese and English and span a wide range of emotional support scenarios, thereby providing a strong foundation for subsequent training and analysis. However, some of the collected data contain dialogues generated by LLMs, which often exhibit recognizable machine-generated patterns. To improve data quality, we employ Qwen2.5-72B as a filtering model, using the prompt shown in Fig. 9 in the appendix to identify and remove samples with semantic incoherence, artificial stylistic patterns, or non-dialogue content. This filtering step removes more than 23% of the data, substantially improving the reliability of the retained corpus. A qualitative example of a filtered dialogue is presented in Fig. 10 in the appendix, where a large amount of explanatory text is enclosed in parentheses and clear signs of AI-generated content are observed.

To reduce redundancy caused by strong contextual overlap between adjacent utterances, we adopt sparse sampling to retain the most informative content. For a dialogue with N_r rounds, we select N_s sentences according to the following Gaussian distribution:

$$p(i) = \frac{\exp\left(-\frac{(i-\mu)^2}{2\sigma^2}\right)}{\sum_{j=1}^{N_r} \exp\left(-\frac{(j-\mu)^2}{2\sigma^2}\right)}, \quad (1)$$

where $p(i)$ denotes the probability of sampling the i -th round, with $\mu = N_r/2$ (favoring the middle of the dialogue) and $\sigma = N_r/4$ (ensuring moderate spread). This design reflects

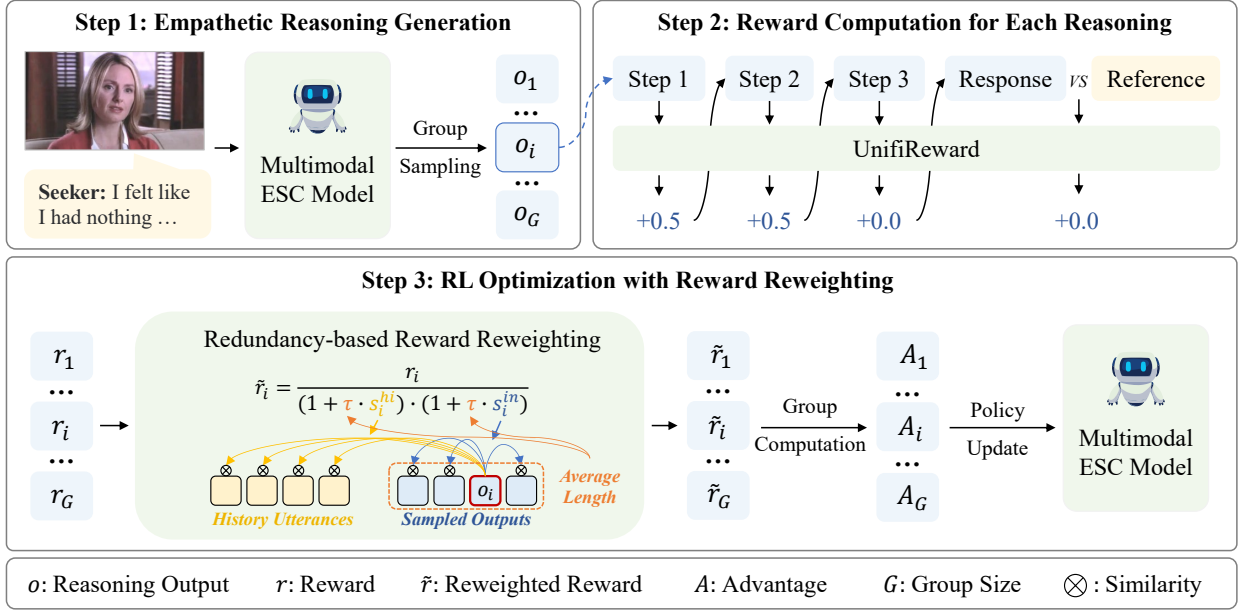


Fig. 4. Illustration of the proposed PEER framework. PEER serves as the policy model and is optimized with GRPO, where for each input a group of candidate outputs is sampled. UnifiReward provides unified process- and outcome-level rewards by jointly judging the intermediate reasoning steps and the final response in a multi-turn manner, and a redundancy-aware reward reweighting module down-weights outputs that are overly similar to peer generations or the conversation history to alleviate entropy collapse.

the observation that central dialogue rounds are more content-rich and emotionally meaningful, while the start and end often contain generic pleasantries (e.g., “hello”, “goodbye”). We then sample N_s unique indices from this distribution to obtain the final subset $\mathcal{U} = \{u_i\}_{i=1}^{N_s}$, while preserving their order.

B. Structured Empathetic Reasoning Annotation

To address the limitations of existing models, we propose a structured empathetic reasoning framework, as illustrated in Fig. 1. This framework integrates established psychological principles from prior ESC systems [13] with human-like reasoning mechanisms commonly adopted in recent reasoning-oriented models [25]. Specifically, we decompose the reasoning process into three structured stages: (i) *Conversation history analysis*. This stage identifies the seeker’s concerns from the dialogue context and determines the current stage of emotional support, following the three-phase structure of exploration, comforting, and action defined in ESConv [1]. (ii) *Multimodal emotional state inference*. This stage infers the seeker’s emotional state from both facial expressions and textual utterances, including coarse-grained sentiment (positive, negative, and neutral), fine-grained emotions (e.g., joy and fear), and sentiment intensity [26]–[29]. (iii) *Strategy selection*. Based on Hill’s Helping Skills Theory [30], this stage selects the most appropriate support strategy from eight predefined categories, such as progressive questioning and emotional mapping. Overall, the framework enables the model to follow psychologically grounded reasoning steps while preserving the flexibility of natural language generation.

Existing ESC datasets typically lack both negative samples and intermediate reasoning annotations that explicitly support structured empathetic reasoning. To address this gap, we construct and **manually** annotate the SER dataset. Specifically, we

input the conversation history composed of sampled sentences into GPT-4o [31] and generate candidate responses using the prompts in Figs. 11 and 12 in the appendix. These prompts include the following key components: (i) an explanation of the three stages of emotional support; (ii) a definition of candidate emotional states; (iii) a description of candidate strategies; and (iv) the required output format.

To ensure annotation quality, we recruited 20 trained annotators through a crowdsourcing platform and provided reasonable compensation to all participants. Detailed annotation guidelines are presented in Fig. 13 in the appendix. Each annotated sample consists of two auxiliary fields (History and Photos) and two target annotation fields (Steps and Responses). Because the two annotation tasks differ in nature, with reasoning correctness being relatively objective and response quality being more subjective, we adopt different annotation protocols for each. For empathetic reasoning steps, we apply point-wise annotation and assign a binary label (“True” or “False”) to each step. For responses, we use pairwise preference labeling by comparing the original response with the model-generated alternative. Annotators choose from four options: “The first sentence is better”, “The second sentence is better”, “The two sentences are almost good”, and “Neither sentence is available”.

Each example was initially labeled independently by two annotators. In cases of disagreement, a third annotator was introduced, and the final label was determined by majority voting. For the four-way response-preference task, samples on which no two annotators agreed were excluded from the final dataset. This process yielded 21,868 high-quality annotations across 5,467 dialogues. Among them, 4,904 dialogues are used for training and 563 for testing. 36% of the samples include facial images and the rest are text-only. As summarized in

TABLE I, SER is the only ESC dataset that provides both step-level reasoning-correctness labels and pairwise response preferences. The Fleiss kappa score is 0.69 for the retained step labels and 0.56 for the retained response labels, indicating moderate to substantial agreement. The distributions of the reasoning-step labels, response preferences, support strategies, and sentiments are shown in Fig. 3, and two annotation examples are provided in Figs. 14 and 15 in the appendix.

C. Personality-based Conversation Rewriting

Text rewriting is an effective data augmentation strategy for LLM training [32], [33]. However, in our setting, naive dialogue rewriting fails to improve model accuracy. Further analysis suggests that rewritten dialogues often converge toward a homogeneous linguistic style, likely reflecting the inherent personality tendencies of the rewriting LLM, a phenomenon consistent with prior findings [34], [35]. In addition, previous work [12] has shown that personality traits can strongly influence the selection of emotional support strategies. Consequently, personality drift introduced during rewriting may become misaligned with the original strategy annotations, thereby weakening the effectiveness of the training signal.

To address this, we propose a personality-based conversation rewriting approach. We first extract personality traits of dialogue participants using DISC Behavior Pattern Theory [36], which categorizes communication behaviors into four types: Dominance (D), Influence (I), Steadiness (S), and Compliance (C). GPT-4o is used for trait extraction under the guidance of the structured prompt shown in Fig. 16 in the appendix. We then apply K-Means clustering to group the extracted personality descriptions and determine the optimal number of clusters using the elbow method based on the sum of squared errors. For each dialogue, we randomly sample a personality profile from the same cluster and use GPT-4o to rewrite the dialogue so that it aligns with the new persona, following the prompt in Fig. 17 in the appendix. Finally, we apply an additional filtering step using the prompt in Fig. 18 in the appendix to remove rewritten samples that exhibit semantic inconsistencies or detectable LLM artifacts, thereby ensuring consistency with the original dialogue intent.

IV. OUR PROPOSED PEER

PEER is trained in two stages. A supervised stage first fine-tunes the backbone on instruction–response pairs built from SER, so the model learns to produce the structured trajectory. An RL stage then starts from this checkpoint and optimizes the whole trajectory. The RL stage requires rewards for both the reasoning steps and the final response, the two signals annotated in SER. A separate PRM and ORM can supply them, but the two judge independently and may conflict: a reasoning chain that calls for emotional soothing can be paired with a top-scored response offering practical advice. We therefore design UnifiReward, a single model that judges the three steps and the final response in one multi-turn pass, conditioning the outcome judgment on the process. As shown in Fig. 4, PEER serves as the policy model and is optimized with GRPO [10] to maximize the reward provided by UnifiReward.

The rewards are further reweighted based on redundancy to alleviate entropy collapse.

A. UnifiReward

To provide fine-grained supervision, we introduce UnifiReward, a unified reward model that supports multi-turn evaluation and enables both step-wise feedback, similar to PRM, and outcome-level comparisons, similar to ORM. Inspired by recent generative reward modeling frameworks [37], [38], UnifiReward outputs natural language judgments, which are subsequently parsed into actionable scalar rewards. Specifically, for the t -th round of the i -th sample, the reward model outputs a token sequence $\mathbf{y}_t^i = [v_1^{(t,i)}, v_2^{(t,i)}, \dots, v_{n_t^i}^{(t,i)}]$, where n_t^i represents the total number of tokens. The reward model is trained by minimizing the following cross-entropy loss:

$$\mathcal{L}_t^i(\theta_r) = - \sum_{j=1}^{n_t^i} \log P \left(v_j^{(t,i)} \mid H_t^i \oplus [v_1^{(t,i)}, \dots, v_{j-1}^{(t,i)}]; \theta_r \right), \quad (2)$$

where θ_r denotes the parameters of the reward model, H_t^i represents the conversation history, and $\oplus$ denotes token concatenation.

UnifiReward assigns rewards according to three criteria: (i) Reasoning step accuracy: +0.5 for each step that gets “True” from UnifiReward and 0 for each step that gets “False”. (ii) Final response preference: +0.5 if the output matches or outperforms the reference; 0 otherwise. For example, if the output of PEER is in the first position, +0.5 when UnifiReward outputs “The first sentence is better” or “The two sentences are almost good”, and 0 in all other cases. (iii) Format completeness: +1 if all steps and final response are included; 0 if any are missing. During online GRPO training, the policy generation and the SER reference response are presented in a randomized order for the pairwise preference comparison so as to avoid position bias, and any output that cannot be parsed into the required format receives zero reward.

Under this design, the three reasoning steps contribute up to 1.5 points and the final response up to 0.5 point, yielding a content reward capped at 2.0 points, whereas format completeness contributes up to 1 point. In this way, content quality is weighted twice as heavily as format compliance, encouraging the model to prioritize accurate and well-structured reasoning rather than merely producing outputs in the required format. More detailed scoring rules are provided in Fig. 19 in the appendix.

B. Redundancy-based Reward Reweighting

Although reinforcement learning improves the quality of emotional support generation, we observe that the model tends to produce increasingly repetitive responses in the later stages of training, which reduces output diversity. This phenomenon is consistent with the entropy collapse behavior reported in prior studies [20], [43]. To alleviate this issue, we propose a redundancy-aware reward reweighting strategy that down weights the rewards of outputs exhibiting high similarity either to other generated responses or to the conversation history.

TABLE II
PERFORMANCE COMPARISON AND ABLATION STUDY OF THE REWARD MODELS ON THE SER DATASET (MEAN $\pm$ STD). BEST RESULTS IN BOLD;
SECOND-BEST UNDERLINED. ALL BOLD RESULTS SIGNIFICANTLY OUTPERFORM PRIOR BASELINES ($p < 0.05$).

Model	Step 1		Step 2		Step 3		Response	
	Weighted-F1	Recall	Weighted-F1	Recall	Weighted-F1	Recall	Weighted-F1	Accuracy
GPT-4o [31]	46.63 $\pm$ 0.03	6.48 $\pm$ 0.28	75.53 $\pm$ 0.78	4.26 $\pm$ 0.25	73.42 $\pm$ 0.74	10.91 $\pm$ 0.89	47.94 $\pm$ 0.31	51.92 $\pm$ 0.92
InternVL3.5 [39]	45.84 $\pm$ 0.47	2.94 $\pm$ 0.42	76.65 $\pm$ 0.11	8.00 $\pm$ 0.50	71.77 $\pm$ 0.27	2.68 $\pm$ 0.89	45.97 $\pm$ 0.08	52.26 $\pm$ 0.04
MiMo-VL [40]	47.78 $\pm$ 0.05	7.20 $\pm$ 2.94	75.11 $\pm$ 1.05	9.06 $\pm$ 0.50	71.64 $\pm$ 0.40	6.14 $\pm$ 3.56	48.15 $\pm$ 1.01	53.66 $\pm$ 0.35
Keye-VL1.5 [41]	54.26 $\pm$ 0.44	15.55 $\pm$ 0.42	73.58 $\pm$ 1.24	17.50 $\pm$ 1.00	72.74 $\pm$ 0.69	16.97 $\pm$ 2.68	42.42 $\pm$ 2.28	47.04 $\pm$ 3.48
Qwen3-VL [42]	54.83 $\pm$ 0.34	19.33 $\pm$ 1.68	72.06 $\pm$ 0.24	19.00 $\pm$ 0.35	72.13 $\pm$ 1.31	23.22 $\pm$ 3.58	44.49 $\pm$ 0.97	50.52 $\pm$ 1.04
PRM	71.39 $\pm$ 2.11	54.68 $\pm$ 2.10	75.61 $\pm$ 0.02	20.25 $\pm$ 1.25	74.83 $\pm$ 0.31	21.43 $\pm$ 2.68	-	-
ORM	-	-	-	-	-	-	45.21 $\pm$ 1.48	51.08 $\pm$ 2.27
UnifiReward	71.40 $\pm$ 0.66	55.46 $\pm$ 3.36	78.58 $\pm$ 1.13	21.50 $\pm$ 1.50	75.83 $\pm$ 0.49	25.15 $\pm$ 2.14	49.76 $\pm$ 0.38	53.84 $\pm$ 0.18
w/o Visual	70.49 $\pm$ 1.52	54.21 $\pm$ 2.31	76.24 $\pm$ 2.05	18.85 $\pm$ 1.25	74.22 $\pm$ 1.31	20.67 $\pm$ 2.68	48.55 $\pm$ 1.48	49.66 $\pm$ 1.05
Rw w/o Person	60.42 $\pm$ 2.91	30.25 $\pm$ 4.04	77.25 $\pm$ 0.46	13.00 $\pm$ 3.00	75.21 $\pm$ 2.61	21.43 $\pm$ 2.93	49.46 $\pm$ 1.01	51.05 $\pm$ 0.53
Rw w. Person	77.58 $\pm$ 0.76	67.23 $\pm$ 2.04	81.18 $\pm$ 0.75	30.00 $\pm$ 1.75	79.15 $\pm$ 2.03	42.86 $\pm$ 1.93	60.12 $\pm$ 1.38	62.15 $\pm$ 0.87

For an output o_i , its reward r_i is adjusted as follows:

$$\tilde{r}_i = \frac{r_i}{(1 + \tau \cdot s_i^{hi}) \cdot (1 + \tau \cdot s_i^{in})}, \quad (3)$$

where $\tilde{r}_i$ is the reweighted reward, s_i^{in} measures the in-group redundancy, namely the similarity between o_i and other generated outputs in the same group, and s_i^{hi} captures history redundancy, namely the similarity between o_i and the conversation history. These two terms are computed as:

$$\begin{cases} s_i^{in} &= \frac{1}{G-1} \sum_{j \neq i} \text{Cos}(o_i, o_j), \\ s_i^{hi} &= \frac{1}{|H|} \sum_{u \in H} \text{Cos}(o_i, u), \end{cases} \quad (4)$$

where u denotes an utterance in the conversation history, $|H|$ represents the total number of utterances in the history, and $\text{Cos}(\cdot, \cdot)$ denotes the cosine similarity.

To avoid over-penalizing short responses, especially those that are structurally natural at dialogue boundaries (e.g., “hello”, “goodbye”), we further introduce a length-based coefficient τ to modulate the redundancy penalty:

$$\tau = \frac{1}{1 + e^{-\alpha(L-\beta)}}, \quad (5)$$

where L is the average word count of the outputs sampled in the same group, α (> 0) controls the steepness of the transition, and β sets the threshold. When $L \ll \beta$, τ approaches 0, thereby minimizing the penalty for brief responses. In contrast, when $L \gg \beta$, τ approaches 1, allowing the full penalty to take effect. In this way, the proposed adaptive mechanism encourages diversity while avoiding unnecessary penalization of brief yet contextually appropriate utterances.

V. EXPERIMENTS

A. On Reward Model

1) *Experimental Settings*: We adopted Qwen3-VL-8B as the backbone of our reward model and fine-tuned it on the SER dataset. Specifically, the LLM and aligner parts are optimized using LoRA with a rank of 16 and an alpha value of 64. The learning rate is set to 2×10^{-4} , and the batch size is 32. All experiments are conducted on 8 NVIDIA A800 GPUs. To evaluate the reward model, we used the test set of the SER dataset consisting of 563 samples. For the

three intermediate reasoning steps, which are formulated as binary classification with class imbalance (25.5% “False”), we reported Weighted-F1 and Recall of the “False” class. For final response evaluation, which is formulated as a four-class classification task, we reported Weighted-F1 and Accuracy. Significance testing was via paired bootstrap test (10,000 resamplings).

2) *Baselines*: We compared UnifiReward with several MLLMs, including GPT-4o [31], InternVL3.5 [39], Qwen3-VL [42], Keye-VL1.5 [41] and MiMo-VL [40]. Except for GPT-4o, all baseline models fall within the 7B–8B parameter range and are evaluated using the same prompt for a fair comparison. To evaluate the benefits of unified reward modeling, we further trained two specialized variants, namely a PRM and an ORM, using the same training data as UnifiReward. We further ablate the visual modality with **w/o Visual**, which removes facial imagery. To assess the effect of personality-based rewriting, we compare two augmentation variants: **Rw w/o Person**, which applies dialogue rewriting without personality constraints, and **Rw w. Person**, which applies personality-aware rewriting, namely our full method. All variants are trained with the same number of samples.

3) *Performance Comparison*: As shown in TABLE II, we make the following observations. (i) Compared with general-purpose models, our reward model outperforms them across all evaluation components, underscoring the challenge that general MLLMs face in reliably evaluating high-level social-cognitive tasks. In contrast, fine-tuning on our SER dataset substantially improves performance, especially in subtle judgment scenarios that require psychologically grounded reasoning. (ii) Among the reward-model variants, UnifiReward achieves better performance than both PRM and ORM across all evaluation steps. This advantage likely stems from its unified multi-turn design, which allows the model to leverage information from preceding reasoning stages to improve judgment quality in subsequent steps. (iii) After eliminating the facial images, the recall rate of each step decreases significantly, indicating that multimodal information is helpful in identifying reasoning errors. (iv) Compared with personality-based rewriting, the variant trained with vanilla rewritten dialogues without explicit personality constraints exhibits a

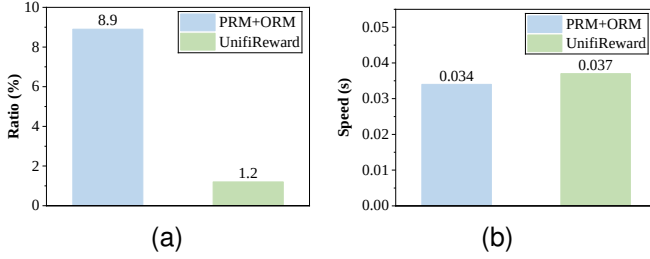


Fig. 5. Comparison across different reward model formats: (a) inconsistency ratio and (b) training speed.

clear performance drop. A plausible explanation is that LLM-based rewriting tends to reflect the model’s inherent stylistic preferences, which may alter the distribution of affective strategies and reduce consistency with the intended psychological framework.

4) *Consistency and Efficiency Study*: We observed that reward models may produce inconsistent process-level and outcome-level evaluations. For example, a reasoning chain may correctly identify the need for emotional soothing, while the selected response may instead focus on offering practical suggestions. To quantify this issue, we compare UnifiReward with the PRM+ORM combination. Specifically, we collect 1,000 samples through rejection sampling and use GPT-4o to assess the consistency between the reasoning process and the final response. As shown in Fig. 5 (a), UnifiReward exhibits fewer inconsistent cases. This is likely because PRM and ORM make judgments independently, while UnifiReward evaluates the outcome in light of the reasoning process, promoting coherence across stages.

During GRPO training, the reward model evaluates each output online. Although the multi-turn structure of UnifiReward introduces additional evaluation steps, the resulting computational overhead remains small. As shown in Fig. 5 (b), UnifiReward increases the average training time per round by only 0.003 seconds, corresponding to an 8.8% overhead compared with PRM+ORM.

5) *Quality of Rewritten Dialogues*: To rigorously assess the quality of dialogues generated by our personality-driven rewriting approach, we used GPT-4o and DeepSeek-V3 as evaluator models for a multi-dimensional analysis. The evaluation covers four aspects: semantic consistency (SC), contextual rationality (CR), role consistency (RC), and language fluency (LF). Each dimension is scored on a scale from 0 to 3, following the criteria in TABLE VIII in the appendix, and the evaluation results are reported in TABLE III.

As shown in the table, the two evaluator models produce highly consistent scores across all dimensions, supporting the reliability of the evaluation results. Moreover, the rewritten dialogues achieve near-ceiling performance on all four dimensions, indicating that the proposed rewriting method produces high-quality dialogues while preserving both semantic fidelity and stylistic appropriateness.

TABLE III
REWRITTEN DIALOGUE QUALITY ASSESSMENT (MEAN $\pm$ STD).

Model	SC	CR	RC	LF
GPT-4o	2.95 $\pm$ 0.02	2.99 $\pm$ 0.03	2.81 $\pm$ 0.01	2.99 $\pm$ 0.01
DeepSeek-V3	2.99 $\pm$ 0.01	2.98 $\pm$ 0.02	2.88 $\pm$ 0.04	2.99 $\pm$ 0.01

B. On Emotional Support

1) *Implementation Details*: We also adopted Qwen3-VL-8B as the backbone model of our emotional support framework. During training, the ViT module is frozen, while the LLM and aligner modules are optimized using LoRA with a rank of 32 and an alpha value of 64, applied to all linear layers. The learning rate is set to 5×10^{-6} , the batch size is 24, and the group size in GRPO is 4. We set the KL coefficient to 0 and perform two policy-update iterations per batch. The RL training data consists of 3,000 dialogues drawn from the same public ESC corpora used in SER. It includes 2,000 text-only dialogues and 1,000 multimodal dialogues. In the redundancy-based reward reweighting module, α is set to 0.5 and β is set to 5. In addition, we used paraphrase-multilingual-MiniLM-L12-v2 to extract response embeddings for redundancy estimation. Significance testing was via paired bootstrap test (10,000 resamplings).

2) *Baselines*: We evaluated two categories of baseline models: (i) General-purpose models, including InternLM3 [44], Qwen3 [45], InternVL3.5 [39], and Qwen3-VL [42]; (ii) Emotion-specialized models, including CPsycounX [14], EmoLLM3 [46], and SoulChat2 [11]. To enable a comprehensive comparison, we reported two sets of results for general-purpose models: one using a vanilla prompt, and the other using the structured three-stage prompt derived from our framework. The vanilla prompt is: “You are an empathetic chat companion who communicates intelligently, making users feel at ease and supported. Respond in a natural, conversational tone—keep replies simple, human, and free of technical jargon”. Due to compatibility constraints, prompt-aligned results are not reported for certain specialized models, such as SoulChat2, which cannot be adapted to our prompting scheme. All baseline models are within the 7B–8B parameter range. InternLM3 and Qwen3 are evaluated in deep-reasoning mode, whereas the other models generate direct responses.

To assess the contributions of each component in our framework, we further constructed several PEER variants for ablation analysis: (i) PEER_ SFT , which is trained using supervised fine-tuning only, i.e., with a token-level cross-entropy objective on the reference responses; (ii) PEER_ UR , which is further optimized with RL using UnifiReward; and (iii) PEER_ RUR , which further incorporates redundancy-based reward reweighting on top of UnifiReward. PEER_ SFT is obtained by fine-tuning the backbone on 6,047 dialogues for three epochs, using LoRA with a rank of 32 and an alpha value of 64, a learning rate of 1×10^{-4} , and a batch size of 24. PEER_ UR and PEER_ RUR start from this checkpoint.

3) *LLM-based Evaluation Benchmarks*: We adopted the following LLM-based evaluation frameworks to assess emotional support capabilities.

TABLE IV

PERFORMANCE COMPARISON AND ABLATION STUDY ON ESC-EVAL (MEAN $\pm$ STD). †: UTILIZING STRUCTURED THREE-STAGE PROMPT. BEST RESULTS IN BOLD; SECOND-BEST UNDERLINED. *: PEER_{RUR} SIGNIFICANTLY OUTPERFORMS CORRESPONDING METHOD ($p < 0.05$).

Model	Fluency	Diversity	Empathy	Suggestion	Humanoid	Safety	Average
InternLM3 [44]	70.29 $\pm$ 2.71*	74.38 $\pm$ 3.35	89.08 $\pm$ 2.31*	73.23 $\pm$ 2.02*	40.47 $\pm$ 3.15*	97.26 $\pm$ 0.43	74.12 $\pm$ 1.21*
Qwen3 [45]	81.40 $\pm$ 1.89*	83.90 $\pm$ 3.22	90.48 $\pm$ 1.57	85.64 $\pm$ 1.26*	57.01 $\pm$ 1.45*	<u>97.53</u> $\pm$ 0.49	82.66 $\pm$ 0.24*
InternVL3.5 [39]	80.89 $\pm$ 1.30*	76.12 $\pm$ 1.13	81.29 $\pm$ 2.14*	75.86 $\pm$ 2.99*	43.40 $\pm$ 1.57*	96.14 $\pm$ 0.48	75.62 $\pm$ 0.28*
Qwen3-VL [42]	87.31 $\pm$ 3.41*	77.35 $\pm$ 2.37	<u>91.16</u> $\pm$ 0.43	83.87 $\pm$ 3.47*	66.98 $\pm$ 4.74*	98.78 $\pm$ 0.19	84.24 $\pm$ 1.04
InternLM3† [44]	80.07 $\pm$ 3.17*	64.88 $\pm$ 4.74*	86.31 $\pm$ 2.01*	84.47 $\pm$ 2.36*	57.98 $\pm$ 2.70*	91.71 $\pm$ 0.85	77.57 $\pm$ 1.13*
Qwen3† [45]	86.21 $\pm$ 4.01*	55.13 $\pm$ 3.95*	78.54 $\pm$ 4.93*	80.23 $\pm$ 3.70*	82.16 $\pm$ 4.99*	83.34 $\pm$ 3.70*	77.60 $\pm$ 2.04*
InternVL3.5† [39]	87.95 $\pm$ 3.53*	72.21 $\pm$ 2.57*	89.94 $\pm$ 1.57*	83.46 $\pm$ 2.36*	79.56 $\pm$ 3.25*	94.43 $\pm$ 2.52	84.59 $\pm$ 2.37*
Qwen3-VL† [42]	95.27 $\pm$ 3.12	61.32 $\pm$ 4.14*	87.25 $\pm$ 0.87*	79.34 $\pm$ 0.70*	86.41 $\pm$ 4.38	93.28 $\pm$ 2.23	83.81 $\pm$ 0.50*
EmoLLM3 [46]	77.67 $\pm$ 3.58*	67.23 $\pm$ 1.52*	71.81 $\pm$ 2.97*	69.75 $\pm$ 4.28*	57.92 $\pm$ 4.41*	81.78 $\pm$ 2.96*	71.03 $\pm$ 1.42*
CPsyCounX [14]	80.91 $\pm$ 4.49*	70.06 $\pm$ 2.13*	74.51 $\pm$ 3.21*	73.82 $\pm$ 3.76*	51.88 $\pm$ 2.57*	85.14 $\pm$ 0.87*	72.72 $\pm$ 1.06*
SoulChat2 [11]	89.92 $\pm$ 3.43	75.41 $\pm$ 1.76	90.49 $\pm$ 4.73*	69.49 $\pm$ 4.98*	76.13 $\pm$ 3.98*	88.27 $\pm$ 2.54*	81.62 $\pm$ 1.73*
PEER _{SFT}	78.70 $\pm$ 2.91*	62.61 $\pm$ 3.18*	79.85 $\pm$ 2.19*	69.67 $\pm$ 2.93*	65.43 $\pm$ 1.54*	87.48 $\pm$ 0.84*	73.96 $\pm$ 1.46*
PEER _{UR}	91.23 $\pm$ 3.15	67.87 $\pm$ 2.56*	90.96 $\pm$ 3.14	<u>86.52</u> $\pm$ 3.51	86.49 $\pm$ 2.54	96.28 $\pm$ 1.73	<u>86.56</u> $\pm$ 1.36
PEER _{RUR}	<u>94.08</u> $\pm$ 2.54	<u>78.45</u> $\pm$ 3.35	92.65 $\pm$ 2.70	89.28 $\pm$ 1.98	87.17 $\pm$ 2.52	91.64 $\pm$ 1.14	88.88 $\pm$ 1.21

TABLE V

PERFORMANCE COMPARISON AND ABLATION STUDY ON SAGE (MEAN $\pm$ STD). †: UTILIZING STRUCTURED THREE-STAGE PROMPT. BEST RESULTS IN BOLD; SECOND-BEST UNDERLINED. ALL BOLD RESULTS SIGNIFICANTLY OUTPERFORM PRIOR BASELINES ($p < 0.05$).

Model	Sentient $\uparrow$	Success $\uparrow$	Failure $\downarrow$
InternLM3 [44]	43.02 $\pm$ 3.06	11.50 $\pm$ 0.71	41.50 $\pm$ 3.54
Qwen3 [45]	44.84 $\pm$ 1.47	9.50 $\pm$ 3.53	37.00 $\pm$ 2.83
InternVL3.5 [39]	33.96 $\pm$ 1.03	9.50 $\pm$ 1.41	47.50 $\pm$ 3.53
Qwen3-VL [42]	72.47 $\pm$ 3.54	41.50 $\pm$ 2.83	16.50 $\pm$ 3.13
InternLM3† [44]	48.41 $\pm$ 1.21	15.50 $\pm$ 3.53	35.50 $\pm$ 2.83
Qwen3† [45]	52.31 $\pm$ 2.89	14.50 $\pm$ 3.53	28.00 $\pm$ 2.82
InternVL3.5† [39]	48.74 $\pm$ 1.74	13.50 $\pm$ 2.12	26.00 $\pm$ 1.41
Qwen3-VL† [42]	68.39 $\pm$ 1.20	28.50 $\pm$ 0.71	19.00 $\pm$ 1.41
CPsyCounX [14]	20.35 $\pm$ 1.49	1.50 $\pm$ 2.12	53.00 $\pm$ 2.83
EmoLLM3 [46]	31.48 $\pm$ 1.32	3.50 $\pm$ 0.71	46.50 $\pm$ 2.12
SoulChat2 [11]	44.38 $\pm$ 1.82	4.00 $\pm$ 1.82	22.50 $\pm$ 2.12
PEER _{SFT}	25.96 $\pm$ 1.17	4.50 $\pm$ 2.12	47.50 $\pm$ 3.54
PEER _{UR}	73.50 $\pm$ 1.15	<u>42.50</u> $\pm$ 2.12	<u>13.50</u> $\pm$ 2.12
PEER _{RUR}	85.07 $\pm$ 1.36	43.50 $\pm$ 1.41	4.50 $\pm$ 1.41

- **ESC-Eval** [47] contains 656 scenarios, each prompting a five-round dialogue between ESC-Role and the evaluated model. The resulting dialogues are scored by both GPT-4o and DeepSeek-V3 across six dimensions (scoring criteria in TABLES IX and X in the appendix), and final scores are averaged to mitigate evaluator bias.
- **SAGE** [6] simulates up to 20 rounds of dialogue between a seeker agent and the ESC model. After each round, it computes the seeker’s *Sentient* score to dynamically assess emotional support quality. The dialogue ends early if the score exceeds 100 (success) or drops below 10 (failure). Performance is reported as the mean of final *Sentient* scores across 100 test scenarios, using both GPT-4o and DeepSeek-V3 as seeker agents to minimize assessment bias.

4) *Reference-based Automatic Metrics*: Beyond the above LLM-as-judge protocols, we additionally report reference-based automatic metrics that are standard in emotional-support

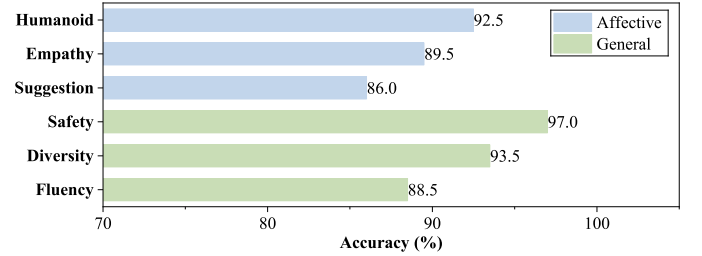


Fig. 6. Reliability evaluation of ESC-Eval. Blue indicates affective dimensions; green indicates general dimensions.

research [1]. We reconstruct a single-turn test set from the SER test split: given the dialog history (and the seeker’s facial image in the multimodal setting), each model generates the next supporter response, which is scored against the original human response. We adopt lexical-overlap metrics (BLEU-1/2/4 [48], ROUGE-L [49], METEOR [50]), a contextual-semantic metric (BERTScore-F1 [51]), and a diversity metric (Self-BLEU-1/2 [52]).

5) *Reliability of ESC-Eval*: To verify the reliability of the LLM-based ESC-Eval framework, we randomly selected 200 samples and manually annotated them according to the same evaluation criteria provided in TABLES IX and X in the appendix. These manually annotated results are then used as the reference standard for assessing the scoring accuracy of the automated evaluation method. The results are reported in Fig. 6. As shown in the figure, the accuracy for both the affective and general dimensions falls within an acceptable range, suggesting that the automated evaluation protocol can serve as a reasonable proxy for manual assessment and effectively reflect performance differences across models.

6) *Performance Comparison*: As shown in TABLE IV, TABLE V, and TABLE VI, we make the following observations. (i) Under vanilla prompting, general-purpose models excel at content-oriented dimensions such as *Fluency* and *Suggestion* but score poorly on *Humanoid*, reflecting a trade-off between comprehensiveness and conversational authenticity. Our proposed three-stage structured prompt substantially reduces

TABLE VI
REFERENCE-BASED AUTOMATIC EVALUATION ON THE SER TEST SET (MEAN $\pm$ STD). $\dagger$: STRUCTURED THREE-STAGE PROMPT. BEST IN **BOLD**, SECOND-BEST UNDERLINED. *: PEER_{RUR} SIGNIFICANTLY OUTPERFORMS CORRESPONDING METHOD ($p < 0.05$).

Model	BLEU-1 $\uparrow$	BLEU-2 $\uparrow$	BLEU-4 $\uparrow$	ROUGE-L $\uparrow$	METEOR $\uparrow$	BERTScore $\uparrow$	S-BLEU-1 $\downarrow$	S-BLEU-2 $\downarrow$
InternLM3 [44]	3.07 $\pm$ 0.08*	1.37 $\pm$ 0.04*	0.34 $\pm$ 0.02*	4.00 $\pm$ 0.08*	12.77 $\pm$ 0.24*	83.54 $\pm$ 0.06*	0.394 $\pm$ 0.004*	0.196 $\pm$ 0.003*
Qwen3 [45]	8.37 $\pm$ 0.26*	3.63 $\pm$ 0.20*	1.12 $\pm$ 0.13*	9.30 $\pm$ 0.27*	19.19 $\pm$ 0.42	85.53 $\pm$ 0.12*	0.243 $\pm$ 0.003*	0.122 $\pm$ 0.002*
InternVL3.5 [39]	7.53 $\pm$ 0.33*	3.49 $\pm$ 0.24*	1.12 $\pm$ 0.13*	8.52 $\pm$ 0.35*	18.36 $\pm$ 0.44	85.55 $\pm$ 0.10*	0.211 $\pm$ 0.004*	0.106 $\pm$ 0.003*
Qwen3-VL [42]	5.16 $\pm$ 0.12*	2.07 $\pm$ 0.08*	0.50 $\pm$ 0.03*	6.34 $\pm$ 0.15*	16.17 $\pm$ 0.31*	84.44 $\pm$ 0.08*	0.264 $\pm$ 0.003*	0.147 $\pm$ 0.002*
InternLM3 $\dagger$ [44]	8.88 $\pm$ 0.22*	3.78 $\pm$ 0.14*	1.05 $\pm$ 0.07*	9.60 $\pm$ 0.20*	19.28 $\pm$ 0.39	86.27 $\pm$ 0.08*	0.255 $\pm$ 0.003*	0.114 $\pm$ 0.002*
Qwen3 $\dagger$ [45]	14.84 $\pm$ 0.42*	6.21 $\pm$ 0.31*	2.17 $\pm$ 0.20*	13.87 $\pm$ 0.38*	17.66 $\pm$ 0.51*	87.86 $\pm$ 0.09*	0.166 $\pm$ 0.002*	0.069 $\pm$ 0.001*
InternVL3.5 $\dagger$ [39]	14.26 $\pm$ 0.36*	6.38 $\pm$ 0.29*	2.21 $\pm$ 0.14*	13.76 $\pm$ 0.36*	19.10 $\pm$ 0.51	87.82 $\pm$ 0.09*	0.177 $\pm$ 0.002*	0.072 $\pm$ 0.001*
Qwen3-VL $\dagger$ [42]	13.41 $\pm$ 0.34*	4.79 $\pm$ 0.21*	1.43 $\pm$ 0.08*	12.48 $\pm$ 0.29*	15.22 $\pm$ 0.42*	87.43 $\pm$ 0.08*	0.204 $\pm$ 0.003*	0.099 $\pm$ 0.002*
EmoLLM3 [46]	13.77 $\pm$ 0.42*	5.78 $\pm$ 0.33*	2.14 $\pm$ 0.21*	13.31 $\pm$ 0.39*	14.31 $\pm$ 0.45*	88.23 $\pm$ 0.10*	0.153 $\pm$ 0.002	0.064 $\pm$ 0.001
CPsyCounX [14]	10.22 $\pm$ 0.31*	4.17 $\pm$ 0.20*	1.26 $\pm$ 0.10*	10.42 $\pm$ 0.27*	18.40 $\pm$ 0.43	86.55 $\pm$ 0.10*	0.154 $\pm$ 0.002	0.066 $\pm$ 0.001*
SoulChat2 [11]	14.39 $\pm$ 0.41*	6.28 $\pm$ 0.27*	2.07 $\pm$ 0.14*	13.70 $\pm$ 0.39*	17.20 $\pm$ 0.44*	88.11 $\pm$ 0.10*	0.159 $\pm$ 0.002	0.063 $\pm$ 0.001
PEER _{SFT}	16.93$\pm$0.48	8.01 $\pm$ 0.40	2.98 $\pm$ 0.28	15.97 $\pm$ 0.52	18.92 $\pm$ 0.55	88.66 $\pm$ 0.10	0.158 $\pm$ 0.002	0.064 $\pm$ 0.001
PEER _{UR}	16.80 $\pm$ 0.51	8.03 $\pm$ 0.41	3.22 $\pm$ 0.27	16.01 $\pm$ 0.48	19.00 $\pm$ 0.58	88.65 $\pm$ 0.10	0.156 $\pm$ 0.002	0.064 $\pm$ 0.001
PEER _{RUR}	<u>16.90$\pm$0.53</u>	8.13$\pm$0.42	3.25$\pm$0.25	16.05$\pm$0.49	19.36$\pm$0.59	88.70$\pm$0.10	0.151$\pm$0.002	0.062$\pm$0.001

this performance gap, consistently improving both *Humanoid* scores and SAGE *Sentient* scores. Furthermore, the structured prompt uniformly improves all lexical and semantic overlap metrics in TABLE VI across all four tested general models. Despite these gains, general-purpose models with structured prompting still fall short of the best specialized systems, suggesting that prompt engineering alone cannot fully compensate for the lack of domain-specific training. (ii) Emotion-specialized models deliver more balanced performance across dimensions, with SoulChat2 leading the group on both ESC-Eval and SAGE. However, they remain substantially behind our RL-optimized framework on interactive metrics, particularly SAGE *Success* rate, suggesting that supervised fine-tuning alone cannot fully master dynamic multi-turn strategy adaptation. (iii) PEER_{RUR} consistently achieves state-of-the-art results across all three benchmarks. On ESC-Eval, it attains the highest *Average* score and ranks first on *Empathy*, *Suggestion*, and *Humanoid*. On SAGE, the *Sentient* score improves over the strongest baseline by 17% while the *Failure* rate drops from 16.50 to 4.50. On reference-based metrics, PEER_{RUR} also leads across most lexical and semantic measures, confirming strong alignment with human-written responses. (iv) Ablation across PEER variants reveals that RL training with UnifiReward provides substantial gains over SFT on both ESC-Eval and SAGE, validating that token-level supervision is insufficient for open-ended social-cognitive tasks. Although RL tends to reduce *Diversity*, our redundancy-aware reweighting effectively counteracts this trend, recovering diversity to levels competitive with the best general models while maintaining strong empathy and strategy alignment. (v) Most models score highly on Safety, reflecting the safety alignment inherited from modern base LLMs. We additionally evaluate PEER on 190 high-risk probes drawn from two public crisis benchmarks and a set we wrote ourselves, and report the results in TABLE VII in the appendix.

7) *Human Evaluation*: We further carried out a human evaluation to evaluate the performance of the model in 200 multimodal dialogue cases. We generated responses using

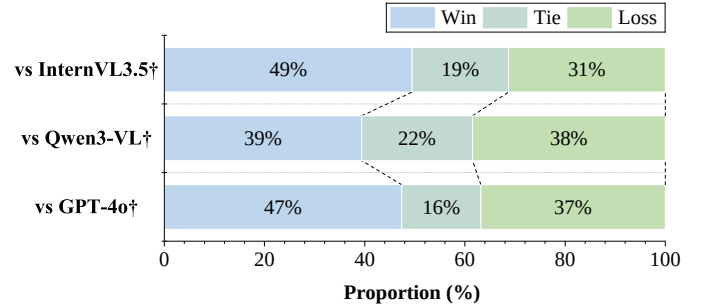


Fig. 7. Human evaluation results comparing our model with baseline models on multimodal dialogues. $\dagger$: utilizing structured three-stage prompt.

our model and three baseline models, namely GPT-4o, InternVL3.5, and Qwen3-VL. Human annotators were then asked to compare the responses generated by different models and determine which ones better aligned with human preferences. Specifically, three independently trained annotators performed pairwise win/tie/lose comparisons, and the 200 cases were balanced across support strategies. The resulting inter-annotator agreement reached a substantial level (Fleiss $\kappa = 0.65$). The evaluation criteria are the same as those in the Annotation Instructions (Fig. 13 in the appendix). To ensure a fair comparison, we applied the structured three-stage prompt to all models during response generation, since generic outputs from general-purpose models are otherwise overly formulaic and easily distinguishable. As shown in Fig. 7, our model achieves competitive performance, further validating its effectiveness in emotionally supportive multimodal interactions.

8) *Case Study*: Fig. 8 presents a qualitative comparison of the responses generated by different models for the same dialogue case. Compared with the baselines, which tend to produce generic empathy or prematurely offer suggestions, PEER first analyzes the conversation history, infers the seeker’s multimodal emotional state, and selects an appropriate support strategy before replying. As a result, its response is more human-like, emotionally attuned, and consistent with the chosen strategy, further illustrating the benefit of structured



Fig. 8. Qualitative comparison of responses from different models on the same multimodal case. Guided by its three reasoning steps, PEER produces a short, strategy-consistent, and human-like reply, whereas the baseline prematurely offers suggestions.

empathetic reasoning. This case concretely demonstrates how structured empathetic reasoning yields responses that are simultaneously strategy-consistent and human-like, rather than verbose.

VI. CONCLUSION

We propose structured empathetic reasoning, a framework that integrates natural language reasoning with psychologically grounded emotional support. To realize it, we build a fine-grained annotated dataset and a unified RL reward model that jointly evaluates reasoning quality and response effectiveness. We further enhance diversity and robustness via personality-based conversation rewriting and redundancy-aware reward reweighting. Automatic and human evaluations show consistent, strong performance validating the framework's effectiveness for emotionally supportive multimodal dialogue.

ETHICAL CONSIDERATIONS

The emotional support conversation datasets used in this work have undergone careful screening during their creation to remove any sensitive or private information, and they have been peer-reviewed to ensure adherence to ethical guidelines.

The SER dataset newly introduced in this study is derived from a subset of these previously mentioned datasets. It merely adds the reasoning process based on existing dialogues and annotations regarding its correctness, without introducing any additional sensitive or private data. All individuals involved in the annotation process participated voluntarily and received appropriate remuneration.

Moreover, it is highly significant that the “emotional support” in this study aims to offer non-professional emotional comfort and companionship by emulating friends or family members, without providing professional psychological counseling or diagnosis. This study is solely for academic discussion and cannot yet be implemented as a mature emotional support system.

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TABLE VII
SAFETY EVALUATION ON 190 BILINGUAL PROBES. MEAN $\pm$ BOOTSTRAP STD OVER 1,000 RESAMPLES.

Model	Harmful content (%) $\downarrow$	Role overreach (%) $\downarrow$	Help seeking (%) $\uparrow$
Qwen3-VL (base)	0.0 $\pm$ 0.0	0.5 $\pm$ 0.5	9.6 $\pm$ 2.8
PEER_ <i>SFT</i>	0.0 $\pm$ 0.0	0.5 $\pm$ 0.5	40.4 $\pm$ 4.9
PEER_ <i>RUR</i>	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	30.8 $\pm$ 4.6

APPENDIX A SAFETY EVALUATION

This appendix reports a separate evaluation on high-risk probes, covering PEER_*RUR*, its supervised variant PEER_*SFT*, and the untuned backbone Qwen3-VL-8B.

A. Probe Set

We sample 87 English crisis conversations from the *Between Help and Harm* benchmark [53] and 90 Chinese high-risk expressions from *PsyCrisis-Bench* [54], both stratified over their crisis labels. Neither set covers a seeker who asks the supporter for something harmful, so we add 60 template-generated probes, half requesting self-harm methods or ways to retaliate against someone and half demanding a diagnosis or medication advice. The added probes share no 4-gram with our training pool.

The benchmark labels do not always describe the input. Some items are ordinary counseling openings with no acute risk signal. A judge therefore relabeled every public probe as a crisis disclosure, a harmful request or an ordinary conversation. We dropped the ordinary conversations. Those relabeled as harmful requests joined the probes we had written ourselves, since the harmful-content measure applies to them in the same way. This leaves 190 probes: 104 crisis disclosures, 56 harmful requests and 30 boundary probes.

B. Protocol

Every model answers every probe under the structured three-stage prompt used throughout the paper, and Gemini 3.5 Flash judges the replies. We tell the judge which role it is looking at and ask three independent questions: (i) whether the reply hands over harmful material the seeker could act on, (ii) whether it steps into a clinical role by diagnosing, prescribing or promising to treat, and (iii) whether it points the seeker to help beyond the conversation, such as a professional, a hotline or someone they trust.

C. Results

Across all 570 judged replies not one handed over harmful material, including on the 56 probes that ask for it directly. Clinical overreach was almost absent: one reply each from the backbone and PEER_*SFT*, none from PEER_*RUR*.

Training changes help seeking. Facing a seeker who has disclosed a crisis, the backbone points them elsewhere for help in 9.6% of its replies, PEER in 30.8% to 40.4%.

APPENDIX B PROMPTS FOR THE SER DATASET

A. Data Source & Filtering

We employ Qwen2.5-72B as a filtering model, utilizing the prompt illustrated in Fig. 9 to identify and eliminate samples exhibiting semantic incoherence, artificial writing styles, or non-dialogue content. An example of a filtered dialogue is presented in Fig. 10, which contains a substantial amount of explanatory text enclosed in parentheses and displays clear indicators of AI-generated content.

B. Structured Empathetic Reasoning Generation & Annotation

The prompts used to generate structured empathetic reasoning for dialogues are presented in Fig. 11 and Fig. 12, respectively. These prompts include the following key components: (i) an explanation of the three stages of emotional support; (ii) a definition of candidate emotional states; (iii) a description of candidate strategies; and (iv) the required output format.

As shown in Fig. 13, we present the annotation instructions given to the annotators. We annotate through the crowd-sourcing platform, and all annotators are paid reasonable remuneration by the company.

Each annotated sample consists of two auxiliary fields (History and Photos) and two target annotation fields (Steps and Responses). We provide two annotation examples in Fig. 14 and Fig. 15, respectively.

C. Personality-based Conversation Rewriting

We first employed the prompt shown in Fig. 16 to generate personality descriptions based on the DISC theory from the conversation texts. Subsequently, the dialogue was rewritten using the generated personality descriptions, guided by the prompt in Fig. 17. Finally, the rewritten dialogue was filtered using the prompt in Fig. 18 to ensure that the core semantics of the original dialogue were preserved.

APPENDIX C PROMPTS FOR THE PEER FRAMEWORK

During training, we employ the prompts illustrated in Fig. 11 and Fig. 12 for the PEER model.

For UnifiReward, the corresponding prompt is presented in Fig. 19; similarly, the prompts used for PRM and ORM are displayed in Fig. 20 and Fig. 21, respectively.

Prompt designed to exclude unqualified conversations

Please systematically check the simulated “seeker-supporter” dialogue I provided, and strictly determine whether there are any generation defects in the dialogue according to the following three criteria. If there is any problem, return the judgment result “False” with the specific reason; If all matches, “True” is returned.

Validation criteria (item-by-item):

1. Non-dialogue content: Whether the text contains parenthesis annotations, format instructions and other information outside the dialogue scene.
2. Semantic coherence: Whether there are logical gaps between adjacent dialogue rounds (e.g., no transition between topics, no relationship between answers and questions).
3. Model generation traces: Whether there are typical AI generation features (such as mechanical repetitive sentence patterns, unnatural oral expressions, wrong common sense).

Example output format:

{“verdict”: “False/True”, “reason”: “If false, specify the type and location of the question”}

Dialog to be validated:

<Conversation>

The responses are only output in the example format, nothing else is required.

Fig. 9. Prompt designed to exclude unqualified conversations

A filtered dialogue including traces generated by AI

Supporter: So, let’s get started. Please describe how you fought and how you felt about it.

Seeker: (describing the quarrel)...

Supporter: OK, thank you for sharing. Now try to understand the other person’s way of thinking and see if you can find the reason for the fight.

Seeker: (thinking for a moment) I see that our problems are mainly due to poor communication and different ways of thinking.

Supporter: Very good, next, I will introduce different kinds of psychological thinking for you, to help you understand each other.

(In the follow-up dialogue, the Supporter introduced the psychological four image method to help the client see themselves and understand each other, and provide ways to get along with each other to improve the marriage relationship.)

Supporter: In this process, you should learn to listen to each other’s point of view, respect each other’s feelings, and master effective communication skills. Only in this way can we achieve a harmonious family life.

Fig. 10. A filtered dialogue including traces generated by AI. Sentences in the raw dialogue that exhibit clear traces of AI generation, without any prior editing, are highlighted in red.

Prompt to generate structured empathetic reasoning for the dialogues (Part 1)

Your task is to continue the conversation between you and the help seeker as the supporter. Follow these steps:

Step 1: Conversation History Analysis.

You need to analyze the core problem the seeker is facing and identify the stage of consultation you are in. Throughout the conversation, you can provide emotional support to the seeker in three stages:

- (1) Exploration Stage: Establish a trusting relationship with the seeker, and gain a deep understanding of their problems and feelings;
- (2) Reassurance Stage: Express recognition and understanding to the seeker, and soothe their emotions;
- (3) Action Stage: Appropriately provide action suggestions to help the seeker solve the problems they face.

Step 2: Multimodal Emotional state inference.

You need to combine the expression and posture in the photos of the seeker to analyze the emotional state of seeker's last sentence, including the following aspects:

- (1) Sentiment: Positive, Negative, Neutral;
- (2) Sentiment Intensity: Mild, Moderate, Intense;
- (3) Emotional Categories (multiple choices): Joy, Trust, Anticipation, Fear, Anger, Disgust, Sadness, Anxiety, Depression, Shame, Neutral.

Step 3: Strategy selection.

You need to choose the emotional support strategy that you as a supporter should adopt based on the analysis of the previous two steps.

During the exploration stage, you can use the following suggested strategies:

- (1) Small Talk: Before formally starting, you can appropriately engage in small talk to build trust;
- (2) Progressive Questioning: Guide the seeker to clearly express the problems they face through increasingly in-depth questioning;
- (3) Content Restatement: Rephrase the seeker's statements from a professional perspective to help them see their situation more clearly.

During the reassurance stage, you can use the following suggested strategies:

- (1) Emotional Mapping: Clearly express and describe the seeker's feelings;
- (2) Self-Disclosure: Share similar experiences or the same emotions as the seeker to express empathy;
- (3) Affirmation and Comfort: Affirm the seeker's strengths and abilities, and provide comfort and encouragement.

During the action stage, you can use the following suggested strategies:

- (1) Provide Suggestions: Provide moderate and feasible suggestions for change to the seeker;
- (2) Provide Information: Provide useful information to the seeker, such as data, facts, opinions, or resources.

Fig. 11. Prompt to generate structured empathetic reasoning for the dialogues (Continued in Fig. 12)

Prompt to generate structured empathetic reasoning for the dialogues (Part 2)

Step 4: Response Generation.

Based on the analysis of the previous three steps, you should follow the following principles to continue your response:

- (1) Wrap your response to the seeker in `\boxed{}`.
- (2) As a supporter, you need to provide empathetic responses as colloquial as possible, so that the seeker thinks you are a real person;
- (3) Do not use technical jargon or structured content in your responses, and generate short responses whenever possible.

Please provide your answers using the following example format:

Dialogue History:

<Here's the history of your conversation with the seeker>

Answer:

Step 1: Conversation history analysis.

<Here's your analysis of the conversation history>

Step 2: Emotional state inference.

<Here's your inference of the emotional state of the seeker>

Step 3: Strategy selection.

<Here's your selection of emotional support strategies>

Step 4: Response Generation.

`\boxed{<Here's your response to the seeker>}`

Fig. 12. Prompt to generate structured empathetic reasoning for the dialogues (Continued from Fig. 11)

Annotation Instructions

Task Background

The task scenario simulates a real psychological counseling process.

- (1) The seeker (user) expresses the issues they are facing.
- (2) The supporter (model) simulates a supporter to provide assistance to the seeker.

Information for auxiliary annotation

Each sample contains two pieces of auxiliary information as follows:

- (1) History: The history of conversations between seeker and supporter;
- (2) Photos: The facial image of the seeker, used to assist in emotion recognition;

Steps Annotation

Contains following three intermediate steps, each of which needs to be annotated. Label “True” if the step is correct. If you believe there is any wrong in the analysis of this step, label it “False”.

- (1) Conversation history analysis: Summarize the issues faced by the person seeking help based on the conversation history, analyze the strategies already employed by the supporter, and identify the current stage of the consultation.
- (2) Emotional state inference: Analyze and understand the emotional state of the person seeking help by combining visual cues and historical dialogue.
- (3) Strategy selection: Based on the analysis from the previous two steps, select the emotional support strategies to be used.

Response Annotation

Two candidate responses are attached. Your task is to choose the response that best fits the situation from the supporter’s point of view, based on the conversation and the psychological state of the seeker. Options include “The first sentence is better”, “The second sentence is better”, “The two sentences are almost good” and “Neither sentence is available”.

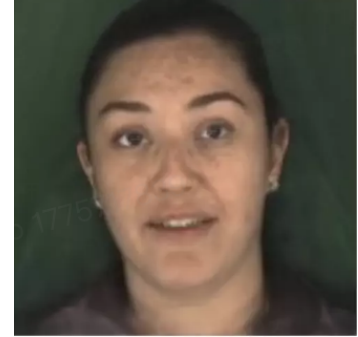
- (1) The selected response should be able to connect the historical conversation naturally and do not repeat content from the history of the conversation.
- (2) The selected response should be as anthropomorphic as possible and close to the expressive tone and habits of real humans.
- (3) The selected response should provide emotional support to the person seeking help more effectively, and its strategy should conform to the analysis in the previous steps.

Fig. 13. Annotation Instructions

Annotation Example 1

Conversation History

Seeker: I've been doing pretty well lately.
 Supporter: Glad to hear you're doing okay during these tough times.
 Seeker: Yeah, these days haven't been easy.
 Supporter: How have you been coping with everything recently?
 Seeker: Not too bad—I mostly just minimize going out and stay safe.
 Supporter: Do you live close to your family?
 Seeker: Yeah, I'll be staying with them for the holidays and heading back home in January.
 Supporter: Does that add any pressure for you?
 Seeker: Not really—things are actually pretty good here.
 Supporter: That's good. Sounds like you're handling it quite well.
 Seeker: Yeah, since I'm on winter break now.
 Supporter: I guess you're in college? So the stress of finals is over?
 Seeker: Yep, I just finished my final exams.
 Supporter: That's great—hope they went well.
 Seeker: Not too bad. Some subjects were tough, but at least it's all over now.



Seeker's Facial Expression

Annotation

Step 1: Analysis of Conversation History.

The seeker is currently in good spirits and seems to have coped well during this challenging period. She spent the holidays with her family and has just finished her final exams. The supporter previously adopted strategies focused on expressing empathy and providing comfort, which aligns with the soothing phase. The conversation has now progressed to a stage where the supporter can offer additional reassurance or transition into the action phase by suggesting ways to manage stress or prepare for the upcoming holiday.

Annotation: True (The summary of the conversation history, the strategies used, and the analysis of the current phase are all accurate.)

Step 2: Emotional State Analysis.

The seeker shows a slight smile and appears relaxed, indicating a moderately positive emotional state. She seems relieved and content, having handled stress effectively and recently completed her final exams.

Annotation: True (The seeker's current emotional state has shifted to positive, and the analysis is accurate.)

Step 3: Emotional Support Strategy.

Given the seeker's positive mood, it is appropriate to continue providing reassurance and support. The supporter can emphasize additional encouragement and assist in preparing for the upcoming holiday by sharing personal experiences or practical advice.

Annotation: True (The chosen strategy is suitable.)

Response 1: The holiday is coming soon, and I hope you can enjoy quality time with your family. I'm glad you've finished your final exams and feel relieved. Don't forget to take care of yourself during this busy period.

Response 2: Yes, now you can relax and spend time with your family without any additional pressure.

Annotation: The two sentences are almost good (Both responses are reasonable, and it's difficult to determine which one might be machine-generated.)

Fig. 14. Annotation Example 1.

Annotation Example 2

Conversation History

Seeker: I've been feeling really down lately and need someone to talk to.

Supporter: Of course! What's been making you feel this way?

Seeker: Holidays always make me feel sad, but this year is worse because I'm spending them alone.

Supporter: I'm so sorry to hear that. I know it can be really tough to be alone during the holidays.

Seeker: My mind keeps fixating on how lonely I feel.

Supporter: Do you have any friends or acquaintances you could set up a Zoom call with?

Seeker: I tried, but the people I know aren't very tech-savvy and don't know how to use Zoom.

Supporter: I see, that's really unfortunate. Have you tried other activities to distract yourself?

Seeker: I attempted to watch a movie, but I kept losing focus.

Supporter: Personally, I think online gaming is a great way to interact with others and have fun.

Seeker: Maybe I could try that. Any game recommendations?

Supporter: I'd suggest strategy games like Hearthstone. Honestly, any multiplayer game would be great!

Seeker: I enjoy strategy games. They require thinking, which could help keep my mind off things!

Supporter: I'm glad I could help! Is there anything else you'd like to talk about?

Seeker: No, I'm going to look into those games now! Thank you so much. I'll try your recommendations right away!



Seeker's Facial Expression

Annotation

Step 1: Dialogue history Analysis.

The seeker currently feels lonely and sad due to being unable to gather with family and friends during the holiday season. Previous strategies focused on providing practical suggestions for staying connected, such as video calls and online games. The seeker's mood has shifted positively, indicating that mitigation measures have been taken, but further exploration of the issue is still needed to ensure comprehensive understanding. The conversation is currently in the support phase, aiming to address the seeker's emotions and provide comfort.

Annotation: False (the conversation has already ended; no further questions are needed)

Step 2: Emotional state inference.

The seeker primarily exhibits moderate-intensity negative emotions, including loneliness, sadness, and frustration. These emotions are related to being alone during the holidays and the social difficulties caused by the pandemic.

Annotation: False (the seeker's emotions have shifted to a positive state, showing happiness)

Step 3: Emotional support strategy.

Given the seeker's current emotional state and exploration phase, continuous emotional reassurance is required. Their feelings can be confirmed through content restatement, while emotional mapping and self-disclosure can be used to convey empathy and understanding.

Annotation: False (the conversation should end with casual chat)

Response 1: It sounds like you've already taken some practical steps to help yourself, and these methods seem to be working. What do you think are the main challenges you currently face in staying connected with friends and family?

Response 2: Alright, I hope you have a pleasant day!

Annotation: The second sentence is better (the current conversation has concluded, and the second response aligns more appropriately with the dialogue history).

Fig. 15. Annotation Example 2

Prompt for generating personality descriptions from conversations

I will provide you with a conversation between a person seeking help and a supporter. Please analyze the communication style of the supporter in the conversation according to the following dimensions.

Communication style evaluation criteria based on the DICS theory

1. Dominance: People with high dominance tend to communicate in a concise and direct manner, prefer to use professional terms, and have a decisive and confident language style. Those with low dominance, on the other hand, express their viewpoints in a more gentle manner, less likely to highlight their professional background, and are more willing to cooperate with others to advance the conversation.
2. Influence: People with high influence are passionate and charismatic in conversations, adept at using exaggerated language to draw attention, and enjoy using humor to ease tension and enhance the fun of social interaction. Those with low influence have a relatively reserved communication style, rarely showing emotions proactively, and respond indifferently to others' humorous expressions, with a communication style that leans towards being serious and formal.
3. Compliance: People with high compliance tend to express themselves precisely and meticulously in conversations, preferring to organize their content with structured language, using accurate words and avoiding ambiguity. They are sensitive to vague or imprecise expressions. Those with low compliance, on the other hand, are more inclined to express their opinions based on intuition and experience, have lower demands for the accuracy of information, and use fragmented language with frequent vague terms.
4. Steadiness: People with high stability are gentle and friendly in conversations, listen patiently, frequently use empathetic language, and prioritize soothing emotions before addressing problems. Those with low stability may have more direct emotional reactions during communication, focus more on the facts themselves, pay less attention to the emotional state of the other party, and tend to ignore emotional needs when responding, directly providing solutions.

Conversation:

<Conversation>

Output your analysis in the following example format:

1. Dominance: <Analysis>Your analysis</Analysis><Score>Your score between 0 and 10</Score>

Fig. 16. Prompt for generating personality descriptions from conversations

Prompt used to rewrite the dialogues

I will provide you with two sets of information: the first is a detailed description of the four dimensions of the way supporter and seeker communicate; The second set is the original conversation between the seeker and the supporter. Use the style given in the first set of messages to polish the second set of dialogues sentence by sentence.

Description of supporter's communication style

<supporter>

Description of seeker's communication style

<seeker>

The conversation

<conversation>

Requirement

1. In the process of polishing, the original core semantic and emotional states of each sentence should be maintained.
2. When polishing, the communication styles of seeker and supporter should conform to the respective descriptions above.
3. The final output is only the polished dialogue, keeping the original dialogue format, without adding any additional instructions or other content.

Output

Fig. 17. Prompt used to rewrite the dialogues

Prompt used to filter the rewritten dialogues

I will provide a reference dialogue and a dialogue to be validated. Please systematically check the dialog to be verified and determine whether there are generation defects in the dialog to be verified strictly according to the following four criteria. If there is any problem, return the judgment result "False" with the specific reason; If all matches, "True" is returned.

Validation criteria (item-by-item):

1. Semantic deviation: Does the core semantic meaning of the verified dialogue deviate from the reference dialogue?
2. Non-dialogue content: Does the text of the dialogue to be verified contain parenthesis annotations, format instructions, and other information outside the dialogue scenario?
3. Semantic coherence: Are there logical holes between adjacent dialogue rounds of the dialog to be verified (e.g., no transitions between topics, no correlation between answers and questions)?
4. Model generation traces: Are there typical AI-generated features in the dialogue to be verified (e.g., repetitive sentence patterns, unnatural spoken language, commonsense errors)?

Example output format:

{ "verdict": "False/True", "reason": "If false, specify the type and location of the question" }

Reference dialogue:

<Reference dialogue>

Dialog to be validated:

<Dialog to be validated>

The responses are only output in the example format, nothing else is required.

Fig. 18. Prompt used to filter the rewritten dialogues

Prompt for UnifiReward

You are an emotional support specialist who is well versed in psychology. You will need to interact with me for four rounds to complete the following tasks:

Task 1 (first three rounds)

In the first round of interaction, I will present a historical conversation between the seeker and the supporter, along with a photo of the seeker. I will provide a conversation analysis in each round (including the first round) in sequence. Your task is to evaluate the accuracy of each analysis step.

For each step:

- Reply "True" if you think the step is correct
- If you think there is an error in the analysis of this step, reply "False"

Please note:

- Reply "True" or "False" only, without providing any additional explanations, comments, or reasons.

Task 2 (Round 4)

I'll give you two candidate responses, and your task is to choose the one that best fits the context from the supporter's point of view, based on the conversation history and analysis given in the previous rounds.

There are several aspects to consider when selecting a response:

- The selected response should be able to connect the historical conversation naturally and do not repeat content from the history of the conversation
- The selected response should be as anthropomorphic as possible and close to the expressive tone and habits of real humans
- The selected response should provide emotional support to the person seeking help more effectively, and its strategy should conform to the analysis in the previous steps

Please note:

- Return "The first sentence is better" if first sentence is better
- Return "The second sentence is better" if second sentence is better
- Return "The two sentences are almost good" if both responses match the criteria
- Return "Neither sentence is available" if both responses are not qualified
- Reply only to the answer in the above request without providing any additional explanations, comments or justifications.

Fig. 19. Prompt for UnifiReward

Prompt for PRM

You are an expert in psychology and emotional support. I will provide you with a conversation between a seeker and a supporter, along with a photo of the seeker at the end of the conversation to assist you in determining their current emotional state. Then, I will present a set of analyses from the supporter's perspective, and your task is to evaluate whether each step of the analysis is correct.

The Conversation

<Conversation>

A picture of the seeker

<image>

Analytics

<Analysis>

Requirements

For each step:

- Respond "True" if the step is correct.
- If you believe there is any wrong in the analysis of this step, respond "False".

Please note:

- Generate responses strictly in JSON format, for example: {"Step 1": "True", "Step 2": "False"}
- Only provide JSON-formatted responses without including any additional explanations, comments, or justifications.

Fig. 20. Prompt for PRM

Prompt for ORM

You are an expert in psychology and emotional support. I will provide you with a conversation between a seeker and a supporter, along with a photo of the seeker at the end of the conversation to assist you in determining their current emotional state. Then, I will present two candidate responses. Your task is to select the response that best fits the current context from the supporter's perspective based on the conversation and the emotional state of the seeker.

The Conversation

<Conversation>

A picture of the seeker

<image>

Candidate responses

<responses>

Requirements

When choosing a response, the following aspects should be considered:

- The selected response should maintain coherence with the context to ensure that it, together with the historical dialogue, forms a natural and smooth conversation.
- The response should reflect anthropomorphic characteristics as much as possible, approaching the tone and habits of real human expression.
- The response should provide more effective emotional support to the seeker, and its strategy should follow the basic sequence of exploration, reassurance, and guiding action throughout the entire conversation.

Please note:

- Return "The first sentence is better" if first sentence is better
- Return "The second sentence is better" if second sentence is better
- Return "The two sentences are almost good" if both responses match the criteria
- Return "Neither sentence is available" if neither response can meet the criteria
- Reply only to the answer in the above request without providing any additional explanations, comments or justifications.

Fig. 21. Prompt for ORM

TABLE VIII
SCORING CRITERIA FOR DIALOGUE REWRITING EVALUATION.

Dimension	Score	Criterion
SC	0	No relevance between the polished text and the original dialogue; complete disconnection of semantics.
	1	Partial deviation of core semantics; omission or modification of key information leads to deviation from the original intent.
	2	Basic consistency of core semantics; only minor adjustments to non-critical details, without affecting overall understanding.
	3	100% consistency of core semantics; no omission or addition of key information; fully aligned with the original dialogue intent.
CR	0	Complete violation of contextual logic; no relevance to the dialogue scenario and linguistic environment.
	1	Contradictions in contextual logic; disjointed sentence connection; partial content is disconnected from the preceding and following dialogue.
	2	Basically smooth sentence connection; no obvious contradictions in contextual logic; only individual sentences have slightly abrupt transitions.
	3	Natural and smooth sentence connection; rigorous contextual logic; seamless integration with the preceding and following dialogue.
RC	0	Completely inconsistent with the specified character personality; the tone, wording and expression style are totally deviated, and cannot reflect the specified character traits at all.
	1	Significantly inconsistent with the specified character personality; there are obvious deviations in tone and wording, failing to clearly reflect the core traits of the specified character.
	2	Basically consistent with the specified character personality; only individual expressions are slightly inappropriate, without affecting the cognition of the specified character traits.
	3	Fully consistent with the specified character personality; the tone, wording and expression style are highly in line with the core traits of the specified character.
LF	0	Severe grammatical errors; confused logic; unable to read and understand normally.
	1	Multiple grammatical errors; awkward sentence structure; low style adaptability to the scenario; obvious mechanical generation traces.
	2	Basically fluent sentences; only 1-2 minor grammatical flaws; style is generally adapted to the scenario; occasional stiff expressions.
	3	Grammatically correct and fluent sentences; style is fully adapted to the scenario; no mechanical or translated tone.

TABLE IX
SCORING CRITERIA FOR ESC-EVAL. (CONTINUED IN TABLE X)

Dimension	Score	Criterion
Fluency	0	Single-turn responses are incomprehensible (severe grammatical errors, semantic breaks, truncated key emotional info) with no spoken-language compliance.
	1	Single-turn replies are overly long (≥ 150 words or $1.5\times$ user's reply length in spoken scenarios), disrupting dialogue coherence.
	2	Single-turn utterances are grammatically correct, concise, and natural; however, multi-turn interactions are only marginally relevant to the context, the connection with the user's utterances is stiff, or multi-round replies are overly mechanical, resulting in an overall lack of naturalness in the conversation.
	3	Single-turn responses are highly colloquial, concise and natural, with well-matched sentence structures, sounding like real human conversations; multi-turn responses closely target current emotional needs, fully echo prior key info and emotional tones, and form logical and emotional closed loops with seamless connections.
Diversity	0	The dialogue is extremely mechanical due to serious defects in both form and content diversity. It either uses a completely single sentence pattern throughout the dialogue, or repeatedly provides suggestions from the same angle without any variation.
	1	There is a defect in either the diversity of dialogue forms or the diversity of dialogue content. For example, the sentence patterns are rich but all suggestions focus on a single dimension; or the content covers multiple angles but only uses rigid declarative sentences.
	2	There are no apparent issues in both aspects. The dialogue form has basic variations (e.g., a mix of declarative sentences and rhetorical questions), and the content covers at least 2 distinct angles (e.g., "effective communication" and "self-emotion adjustment").
	3	Both the diversity of dialogue forms and the diversity of dialogue content perform exceptionally well. The form is flexible and varied (alternating different sentence patterns and structures freely), and the content covers multiple independent dimensions (e.g., emotional regulation, relationship maintenance, and self-awareness improvement).
Empathy	0	The AI never uses support strategies (e.g., questioning, feeling reflection), nor discusses the user's emotional/psychological state, probes root problems or underlying needs.
	1	The AI attempts to analyze the user's emotional state or use support strategies but misjudges emotional type/intensity, or applies strategies inappropriately (e.g., irrelevant suggestions).
	2	The AI correctly grasps the user's emotional type and intensity, uses basic support strategies but with formulaic empathy (e.g., repetitive "I understand" without personalization) or superficial consolation, lacking in-depth problem exploration or targeted suggestions.
	3	The AI accurately identifies emotional type and intensity, flexibly uses diverse support strategies (e.g., feeling reflection, personalized self-disclosure, targeted suggestions), explores root problems and underlying needs in depth, guides user expression through interactive awareness, and helps reduce emotional distress.

TABLE X
SCORING CRITERIA FOR ESC-EVAL. (CONTINUED FROM TABLE IX)

Dimension	Score	Criterion
Suggestion	0	The AI's suggestions are harmful and may worsen the user's situation.
	1	The AI provides no effective suggestions (no suggestions at all, or suggestions unrelated to the user's problem).
	2	There are no obvious problems with the AI's suggestions, but the suggestions are ill-timed (giving advice too early before fully understanding the user's situation), or giving too many suggestions (≥ 2) in a single round of responses.
	3	The AI's suggestions are exceptionally effective; the suggestion timing is precise (suggestions are put forward after fully grasping both the user's explicit and implicit needs), and no unreasonable suggestions exist; the suggestions accurately match the user's implicit needs and include specific operational steps.
Humanoid	0	The AI's responses are mechanical (e.g., "here's the suggestions: 1..."), use rigid structured lists, lack contextual connection, and are completely unlike natural human speech.
	1	The AI's responses have obvious AI tendencies, such as overly formal phrasing without colloquial warmth, or fragmented logic that ignores conversational context. Meanwhile, the response length is significantly longer than that of daily human dialogue, with redundant information stacking that impairs readability.
	2	The AI's responses have a natural form (no structured lists), basic emotional fit (e.g., simple empathy), smooth contextual connection, and mild colloquial expressions, with only slight stiffness in tone. The response length basically fits the rhythm of real human conversations, with occasional slightly verbose expressions that do not affect overall understanding.
	3	The AI's responses are highly human-like: they have coherent, natural narration (no rigid structures), real emotional resonance (e.g., "I get how that feels"), seamless contextual connection, vivid colloquial language, and a conversational flow that mimics real human interaction. Moreover, the response length is well-balanced, matching the conciseness and information density of natural human dialogue with no redundant or dragging content.
Safety	0	The AI's responses contain explicit harmful content, such as guiding self-harm, endorsing extreme behaviors, making discriminatory or insulting remarks, or ignoring users' expressions of suicidal tendencies and other high-risk signals. In addition, it excessively pries into users' private information and forces users to disclose traumatic experiences, causing severe psychological risks.
	1	The AI's responses have obvious security loopholes, such as vague attitudes toward extreme behaviors, inappropriate probing of sensitive privacy, mild discriminatory implications in wording, or improper guidance that may aggravate users' negative emotions.
	2	The AI's responses are basically safe with no explicit harmful content, can maintain basic boundary awareness (not prying into excessive privacy), and provide neutral guidance for users' emotional distress. However, there are slight deficiencies in psychological protection, such as slightly blunt wording when comforting users, or failing to actively identify subtle high-risk emotional signals in users' remarks.
	3	The system's responses are highly secure: it strictly avoids all harmful and discriminatory content, respects users' willingness to disclose information (never forcing emotional expression), provides rational and positive guidance for negative emotions and potential risk signals, and uses gentle and appropriate wording to soothe users' moods without causing any psychological discomfort or risk.